# Pranav Reddy

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## PROJECTS & EXPERIENCE

**Quantum Codesign Lab** — **June 2025 — present**

*Niu Group, UCSB Undergraduate Researcher*

- Designed time-dependent Lyapunov control schedules with learnable parameters to accelerate ground state convergence in the Transverse Field Ising Model (TFIM) via feedback-based quantum algorithms (FQAs)
- Benchmarked GRU, Bayesian, and threshold decoders on continuous stabilizer readout across four hardware-realistic simulation phases; GRU achieved 96% accuracy under colored noise, measurement backaction, and drift

**Pando.AI** — **July 2024–January 2025**

*Engineer Intern* — *https://pando.ai*

- Built a real time retrieval-augmented automation pipeline with LLM-based inference to supply chain workflows
- Developed scalable alerting infrastructure and orchestration logic to improve response reliability and operational throughput in production enterprise environments

**TensorDescent** — **January 2026 - April 2026**

*Language: Python*

- Built multi-stage ML compiler pipeline using PyTorch FX by implementing ShapeProp, DCE, and operator fusion passes, reducing op count and eliminating intermediate memory roundtrips.
- Designed a lifetime-based memory planner for on-chip buffer reuse, mirroring memory tradeoff constraints

**KernelForge** — **February 2026 - April 2026**

*Language: Python*

- Built a Triton-inspired kernel compiler that lowers tensor ops through loop tiling (32x32) and CUDA-style codegen, mapping work across GPU thread hierarchies via block/threading indexing
- Implemented fusion and tiling optimization passes that eliminate kernel launch overhead and keep intermediate tensors in shared memory rather than global DRAM

**AccelSim** — **February 2026 - April 2026**

*Language: Python*

- Built a cycle accurate neural accelerator simulator modeling 5-instruction ISA with configurable latency, on-chip buffer management, and memory traffic tracking
- Implemented bottleneck detection classifying workloads as compute-bound, memory-bound, or buffer limited

**Real-Time Quantum Error Decoding (3-Qubit Bit-Flip Code)** — **January 2026–Current**

*Language: Python*

- Simulated continuous noisy measurement signals from a 3-qubit error-detecting code, modeling hardware realistic error processes for fault-tolerant quantum systems
- Developed decoding pipelines to identify and correct bit-flip events in real time, benchmarking learned models against optimal Bayesian inference baselines

- Engineered scalable experimental infrastructure to evaluate performance across stochastic noise regimes, emphasizing latency, reliability, and hardware-constrained signal decoding
- Explored sequence based and graph based ML approaches (RNNs, GNNs) to improve error classification accuracy under compute constraints

**Application of Graphs to Machine Learning** **January 2024–March 2024**

*Language: C++*

- Implemented graph algorithms (BFS, DFS) to traverse and manage neural network structures, developed methods for node and connection management and applied graph traversal algorithms for prediction and back-propagation

## EDUCATION

**BS Electrical and Computer Engineering | University of California, Santa Barbara** **September 2023 — Present**

Pursuing electrical engineering major with an emphasis on Photonics, ML System Accelerators, High Performance Computing, Computer Architecture, Quantum Error Correction, ML/AI optimizations.

## COURSE WORK

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|---|---|---|
| Intro to Photonics (ECE136A) | Probability and Stats (ECE139) | Computer Architecture (ECE152) |
| Quantum Photonics (ECE 136C) | Intro to Fields & Waves (ECE134) | Fiber Optic Systems (ECE 135) |
| Intro Solid State Devices (ECE132) | Intro to Optimizations (ECE 133) | Digital Circuit Design (ECE 10) |
| Problem Solving in C++ (CS 24) | Signal Processing (ECE 130A) | Discrete Processing (ECE 130B) |
| Interface Design (ECE153B) | Digital Design (ECE152) | Waveguide Physics (ECE 144) |

## SKILLS & LEADERSHIP

**Languages:** C++, C, Python, Java, Arduino, LATEX, SystemVerilog, low-level Linux

**Tools:** Unix, Git/Github, LangChain, Tavily, OpenAI, Claude, Gemini, Quartus, Logisim, FPGA

**Sigma Phi Epsilon** - Vice President / Housing Manager / Siblinghood Chair September 2023 - present